

Table of Contents

1. What is Git	1
Users.....	2
2. Git and GitHub.....	2

1. What is Git

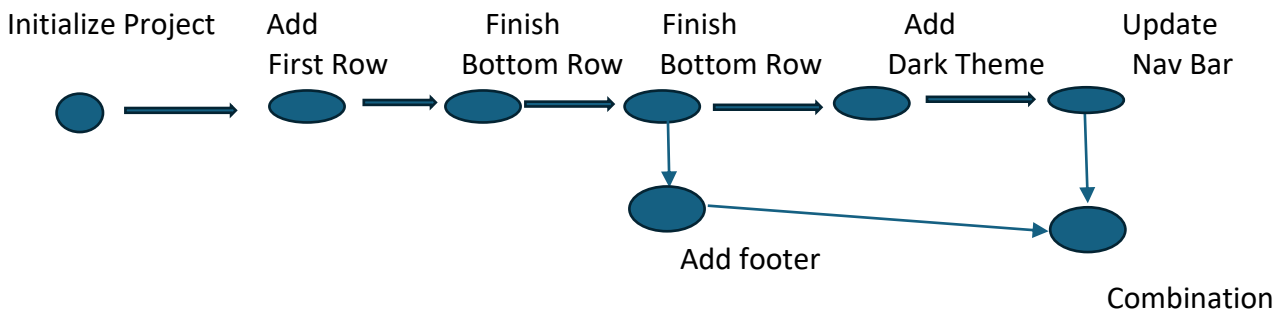
Git The world's most popular version control system. Version Control Version control is software that tracks and manages changes to files over time. Version control systems generally allow users to revisit earlier versions of the files, compare changes between versions, undo changes, and a whole lot more.

What exactly does Git do for us?

1. Track changes across multiple files
2. Compare versions of a project
3. "Time travel" back to old versions
4. Collaborate and
5. Share changes
6. Combine changes
7. Git helps us Revert to a previous version

Example:

Assume you start a New Project. Each step developer of website saves their work and give name. Since previous version is saved, user can anytime "Time travel" meaning go to any saved changed. However, if other team member is working on other feature like designing Home page or adding footer. Their work can be combined later.



Other Version control rules

Git is just one of the many version control systems available today. Other well-known ones include Subversion, CVS, and Mercurial. They all have similar goals but vary significantly in how they achieve those goals. Fortunately, we only need to care about Git because Git is the clear "winner". According to survey nearly 90% of respondents reported Git as their version control system of choice. Over the last few years, the survey hasn't even bothered to ask about version control because Git is so widely used.

Git	– 88.4%
Sub version	- 16.6 %
Team foundation Version Control	– 11.3%
Copying and pasting files to network shares	– 7.7%
Zip file back-ups	– 7.7%
Mercurial	– 3.7%

Linus Torvalds is a legendary software engineer. He is the creator and main developer behind Linux and Git! In 2005, while working on Linux, he became frustrated with the available version control systems. The existing tools were slow, closed source, and usually paid. Torvalds wanted a version control system that was super-fast AND free, unlike the existing tools. On April 3rd, 2005, he got to work on his own VCS, which would become Git. In a matter of days, he had most basic functionality done. The first official Git release came a couple months later. 15 years later in 2020, over 90% of developers worldwide use Git daily.

Users

Engineers & Coders: From massive tech giants like Facebook and Google to the tiniest of startups, developers across the globe use Git. If you plan on becoming a developer, Git is essentially a must-have.

Tech-Adjacent Roles: Many people in non-developer roles end up learning the basics of Git to collaborate with their coworkers. Designers often need to work with Git.

Governments: In recent years, governments have started using Git to manage the drafting of laws. Washington DC City council uses Git & Github to publish its laws. Citizens have even found and fixed typos collaboratively using Git!

Scientists: Git is commonly used by research teams at universities and agencies around the world to manage code, data sets, and more. Git + Github is especially powerful in the peer-reviewed world of science and research.

Writers: Some writers love using Git to manage drafts of complex novels, screenplays, or other works with lots of moving parts and constant changes across multiple files. Git is gaining popularity for use in collaborative textbook writing with multiple authors.

Anyone Really: People find very creative uses for Git ranging from keeping a daily diary to drafting PhD theses to tracking changes to photoshop files. At least one composer writes his symphonies using Git!

2. Git and GitHub

Github is a hosting platform for Git repositories. You can put your own Git repos on Github and access them from anywhere and share them with people around the world.

Beyond hosting repos, Github also provides additional collaboration features that are not native to git. (but super useful). Basically, Github helps people share and collaborate on repos.

Github gives your code a home in the cloud. Where you can record and rewind any change to your code. Host it all for free with unlimited public and private repositories.

Today Github is the home of open-source projects on the internet.

Your Github profile showcases your own projects and contributions to others' projects. It can act as a sort of resume that many employers will consult in the hiring process. Additionally, you can gain some clout on the platform for creating or contributing to popular projects.

Github is not only options, others like GitLab, Gerrit, Bitbucket does the same. Github founded in 2008, Github is now the world's largest host of source code. In early 2020, github reported having over 40 million users and over 190 million repositories on the platform.

Github offers its basic services for free! While Github does offer paid Team and Enterprise tiers, the basic Free tier allows for unlimited public and private repos and collaborations.

www.github.com/pricing

Git	Github
Git is the version control software that runs locally on your machine. You don't need to register for an account. You don't need the internet to use it. You can use Git without ever touching Github	Github is a service that hosts Git repositories in the cloud and makes it easier to collaborate with other people. You do need to sign up for an account to use Github. It's an online place to share work that is done using Git.

From the previous example, all these are all on local machines. What if the local machine is lost or goes bad. All work is lost. I can upload my history to cloud and recover from the cloud. Using Github does the same job. The other advantage is that user/owner of work can share with others in the team as well.

